

Intelligent Circular Economy: Machine Learning-Driven Customer Retention and Dynamic Recommendation in Durable Appliance Leasing Platforms

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Academic and strategic research artifact for the AI-Powered Household Appliance Rental System (RentAI / SDC2). Demonstrating circular economy transition, machine learning churn mitigation, FunkSVD recommendation, and localized generative AI edge deployment.

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Abstract

The rapid acceleration of urban labor mobility has precipitated severe market friction in household durable goods acquisition. Traditional retail purchase models enforce prohibitive upfront capital constraints and promote premature electronic waste (e-waste) generation when tenants relocate. Conversely, durable appliance subscription leasing embodies a core Product-Service System (PSS) paradigm within the Circular Economy. However, subscription ecosystems face heightened vulnerability to customer attrition and choice overload. This paper presents **RentAI**, an integrated architectural framework combining pure NoSQL document persistence, a 15-feature LightGBM classification engine for preemptive churn mitigation, a tri-paradigm collaborative filtering recommendation system, and local privacy-preserving Retrieval-Augmented Generation (RAG). Evaluated across 1,433 catalog appliances and empirical tenant cohorts, our LightGBM churn pipeline achieves an unprecedented **ROC-AUC of 0.9893** and **95.67% test accuracy**, while biased FunkSVD matrix factorization secures an **RMSE of 1.2015** with 100% catalog coverage. Furthermore, local inference offloading via quantized edge models eliminates recurrent cloud API token expenditures (\$0.00 marginal cost) while preserving complete user privacy.

Keywords: Circular Economy, Product-Service Systems, Customer Churn, LightGBM, Collaborative Filtering, FunkSVD, Local RAG, MongoDB.

1. Introduction

Transitioning from a linear "take-make-dispose" economy toward a regenerative circular economy is imperative for sustainable resource utilization (Geissdoerfer et al., 2017). Durable household appliances—encompassing refrigeration systems, washing machines, and HVAC climate control units—exhibit substantial embedded embodied energy and critical raw materials. Despite possessing operational lifespans exceeding 8 to 12 years, appliances owned by mobile urban populations are frequently discarded or liquidated prematurely due to high transit friction during residential relocations (Tukker, 2004).

While the Product-as-a-Service (PaaS) subscription model aligns economic incentives with prolonged asset longevity, appliance rental platforms encounter two critical operational bottlenecks:

- 1. Asymmetric Churn Dynamics:** Subscription margins rely on prolonged lease tenures; unanticipated early contract terminations destroy operational profitability.
- 2. Information Overload & Cold-Start Friction:** Prospective tenants struggle to identify compatible appliance dimensions, power constraints, and package pricing without personalized recommendation systems.

2. Proposed Architecture & Methodology

2.1 Behavioral Churn Prediction Pipeline

We formulate customer churn as a binary classification problem. Rather than relying solely on demographic attributes, we engineer 15 behavioral telemetry features spanning lease tenure, inactivity ratios, payment delinquency, ticket dissatisfaction, and cart friction. We employ LightGBM (Ke et al., 2017), utilizing Gradient-Based One-Side Sampling (GOSS) and Exclusive Feature Bundling (EFB):

$$\text{Loss}(y, F(x)) = - \sum [y_i * \log(p_i) + (1 - y_i) * \log(1 - p_i)]$$

2.2 Tri-Paradigm Recommendation System

To overcome sparsity in implicit rental interactions, we deploy a biased FunkSVD matrix factorization pipeline:

$$\hat{r}(u, i) = \mu + b_u + b_i + p_u^T * q_i$$

where μ denotes the global mean rating, b_u represents user rating bias, b_i denotes appliance affinity bias, and $\mathbf{p}_u, \mathbf{q}_i \in \mathbb{R}^{20}$ represent user and item latent factor projections.

3. Experimental Results & Benchmarks

Evaluation Metric	LightGBM Churn Engine	Baseline Random Forest	Baseline Logistic Reg.
Test Accuracy	95.67%	89.41%	78.23%
ROC-AUC	0.9893	0.9412	0.8125
Precision	88.24%	81.15%	69.40%
Recall	92.31%	83.70%	71.20%
F1-Score	90.23%	82.40%	70.28%
Inference Latency	4.8 ms	18.4 ms	2.1 ms

4. Conclusion

The RentAI platform demonstrates that combining high-performance document stores (MongoDB) with gradient-boosted behavioral models and edge-hosted generative AI yields a sustainable, low-latency, and economically viable foundation for circular economy durable leasing.